

Johnathon Zheng

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EDUCATION

University of California, Berkeley

B.A., Applied Mathematics; Minor in Computer Science

Berkeley, CA

Aug. 2025 – May 2027

Skyline College

A.S., Computer Science & Mathematics

San Bruno, CA

Aug. 2023 – May 2025

Coursework: Artificial Intelligence, Discrete Math, Data Structures & Algorithms, Data Science, Databases, Computer Architecture, Ordinary Differential Equations, Linear Algebra

WORK EXPERIENCE

Peer Tutor

Skyline College

Jan 2025 – May 2025

San Bruno, CA

- Supported Skyline College students in Calculus I-III through weekly 1:1 and drop-in tutoring sessions.
- Helped students develop problem-solving strategies and study habits for success in STEM courses.

Teaching Assistant

KTBYTE Computer Science Academy

Jun 2023 – Sep 2023

Remote

- Conducted regular 1:1 office hours to offer support with Java-related concepts and problem sets.
- Adapted teaching methods to accommodate students of varying ages and backgrounds.

RELEVANT ACADEMIC EXPERIENCE

Worldcraft – Image Reconstruction in Minecraft | *Python, Bash, Linux*

Sept 2025 – Present

- Collaborating on the design of a 3D vision pipeline to reconstruct parts of the UC Berkeley campus in Minecraft using neural radiance fields (NeRFs).
- Prototyping scripts to organize image and semantic segmentation datasets and configure Nerfstudio-based training runs on available GPU resources.
- Exploring approaches to export NeRF outputs into semantic point clouds and voxel grids suitable for Minecraft schematics.

Garbage Classification with YOLO11 | *Python, Ultralytics, Jupyter, Kaggle*

Oct 2024 – Nov 2024

- Fine-tuned a YOLO11 classification model on a 3-class garbage dataset (trash, metal, paper) using the Ultralytics API on Kaggle.
- Cleaned and reorganized raw image data into class folders and created train/validation/test splits with reproducible scripts.
- Implemented training and validation in Jupyter Notebooks, monitoring classification accuracy to evaluate model performance.

Home Server / Homelab | *Proxmox, Docker, Ubuntu Linux, Networking*

May 2023 – Present

- Managed a Proxmox cluster across multiple nodes, running VMs and containers for storage, media, and experimentation.
- Improved reliability by configuring ZFS pools with redundant disks, preventing data loss during testing and upgrades.
- Ran a TrueNAS SCALE VM for centralized storage, SMB sharing, and automated backups.

Timely Track @ SF Hacks | *Python, Flask, HTML/CSS*

Apr 2024

- Built a scheduling assistant that uses Google Maps APIs to ensure students arrive on time to appointments and classes.
- Integrated OpenAI APIs in a Flask backend to reason about which meetings could be replaced with emails or async updates.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML/CSS, Bash

Frameworks & Libraries: Flask, Jinja2, OpenCV, Ultralytics YOLO

Developer Tools: Git, Docker, Proxmox, Linux, TrueNAS, Tailscale

Data & ML: Jupyter Notebooks, NumPy, Pandas, Kaggle